

EX.NO: 10

IMPLEMENTATION OF K-MEDOIDS CLUSTERING ALGORITHM

AIM:

To group the customers of a bank based on their credit card details into three different clusters with the help k-medoids clustering algorithm.

PROCEDURE:

1. Install the necessary packages in R Studio such as “cluster” and “factoextra”
2. Set the current working directory
3. Save the Credit Card details dataset in the current working directory
4. Read the dataset in the form of csv file using **read.csv ()**
5. Finally create the model (k=3) with the help of k-medoids clustering algorithm

SOURCE CODE:

```
install.packages("factoextra")  
  
library(factoextra)  
  
#load data  
df <- USArrests  
  
#remove rows with missing values  
df <- na.omit(df)  
  
  
  
  
  
  
  
  
  
#scale each variable to have a mean of 0 and sd of
```

```
1 df <- scale(df)
```

```
#view first six rows of dataset
```

```
head(df)
```

```
fviz_nbclust(df, pam, method = "wss")
```

```
#calculate gap statistic based on number of clusters
```

```
gap_stat <- clusGap(df, FUN = pam, K.max = 10, #max clusters to consider B = 50)
```

```
#total bootstrapped iterations #plot number of clusters vs. gap statistic
```

```
fviz_gap_stat(gap_stat)
```

```
#make this example reproducible
```

```
set.seed(1)
```

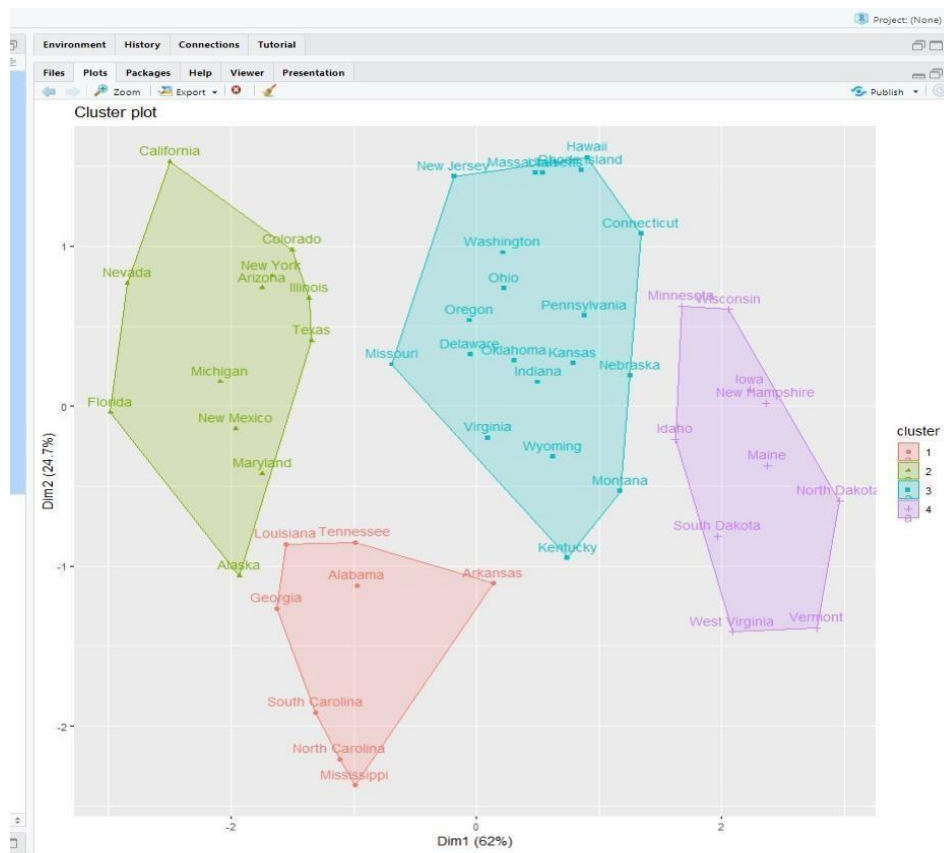
```
#perform k-medoids clustering with k = 4
```

```
clusters kmed <- pam(df, k = 4)
```

```
#view results kmed
```

```
#plot results of final k-medoids model fviz_cluster(kmed, data = df)
```

OUTPUT:



RESULT:

Thus, the three different clusters are created with the help of k-medoids clustering algorithm.